^{15}O - Comments on evaluation of decay data

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This evaluation has been completed with the literature available by February 2026. The former evaluation was completed in 2002.

This radionuclide is one of the T_z=1/2 mirror β transitions surveyed in order to test the predictions of the Standard Model, and to search for possible signals of new physics [2023Se01].

1 Nuclear data

The ground state of ^{15}O decays at 100% by EC/β^+ emission to the ground state of ^{15}N. Indeed, the energy of the first excited state of ^{15}N is 5270.155 (14) keV [1991Aj01], above the adopted Q-value of 2754.18 (49) keV from AME2020 [2021Wa16].

The spins and parities of these two ground states have been adopted from [1991Aj01]: both are 1/2$^-$. The ground state of ^{15}N is stable.

1.1. Half-life

Table 1 summarizes the measured half-lives of ^{15}O decay available in the literature. Additional measurements have been found but not considered. The value reported in [1935Mc01] corresponds to the measurement reported in [1934Li04]. The value from [1957Ki22] seems to come from an average between their measurement and the values from [1950Br29, 1954Kl36]. In addition, it is superseded by [1959Ki99].

The whole dataset is discrepant (χ^2 = 7.53 and $\chi^2_{crit.}$ = 2.25) but the measurements are disparate in quality. The most recent value from [2020Bu02] is the most precise. The values from [1950Br29, 1963Va31] are excluded by Chauvenet criterion. The dataset is discrepant due to the measurements from [1957Pe12, 1959Ki99], which are both scarcely described. These two values have thus been excluded.

All the values with an uncertainty larger than ten times the uncertainty reported in [2020Bu02] have been excluded, i.e., 0.5 s. The resulting dataset is composed of the values from [1960Ja12, 1977Az01, 2020Bu02] and is consistent (χ^2 = 1.76 and $\chi^2_{crit.}$ = 4.61). The adopted value of the ^{15}O half-life is the weighted average with minimum experimental uncertainty, i.e., 122.266 (49) s or 2.0378 (8) min. The value from [2020Bu02] weighs 78%.

Table 1 – Measured half-lives of ^{15}O decay.

Reference	$T_{1/2}$ (s)	Comment
1934Li04	126 (5)	Uncertainty too large; excluded.
1950Br29	118 (6)	Uncertainty too large; excluded.
1954Kl36	123.4 (13)	Uncertainty too large; excluded.
1955Ba83	121 (3)	Uncertainty too large; excluded.
1957Pe12	123.95 (50)	Uncertainty too large; excluded.
1959Ki99	124.1 (5)	Uncertainty too large; excluded.
1960Ja12	122.1 (1)	
1963Cs02	125 (2)	Uncertainty too large; excluded.
1963Ne05	122.6 (10)	Uncertainty too large; excluded.
1963Va31	114 (12)	Uncertainty too large; excluded.
1977Az01	122.23 (23)	
2020Bu02	122.308 (49)	
Adopted	**122.266 (49)**	**Weighted mean with minimum experimental uncertainty**

1.2. Beta transition

The ground-state-to-ground-state EC/β^+ transition is of allowed nature. The BetaShape program [2019Mo35, 2023Mo21] has been used to determine the average energy of the continuous spectrum, the fractional atomic shell electron-capture probabilities, the EC/β^+ ratios and the log *ft* value. The EC/β^+ splitting of the branch has been deduced as $P_{0,0}$(EC) = 0.0999 (17) % and $P_{0,0}(\beta^+)$ = 99.9001 (17) %.

The average energy of the β^+ spectrum has been calculated as E_{av} = 733.47 (23) keV. $P_{0,0}$(EC) and $P_{0,0}(\beta^+)$ have been determined from the EC/β^+ ratio of 0.001000 (17). This is in excellent agreement with the $P_{0,0}$(EC) value of 0.100% given in [2023Se01], which was calculated from tabulated values in [1977Ba48]. The K/β^+ ratio from BetaShape is 0.000928 (16), in good agreement with the measured value from [1972Le33], K/β^+ = 0.00107 (6). The log *ft* value has been determined to be 3.6449 (6), in excellent agreement with the calculated value of 3.6446 (6) from [2023Se01] when including the radiative correction.

Several measurements of the maximum energy of the β^+ continuous spectrum were reported in the literature. They are summarized in Table 2. The dataset is largely discrepant (χ^2 = 38.36 and $\chi^2_{crit.}$ = 3.02) due to the values from [1937Step, 1950Br29] which were clearly underestimated. That can easily be explained by the thicknesses of the material through which the β particles passed, a phenomenon difficult to correct accurately. Therefore, these two values have been excluded. The resulting dataset is consistent (χ^2 = 0.96 and $\chi^2_{crit.}$ = 3.79). The evaluated maximum energy corresponds to the weighted average with minimum experimental uncertainty: E_{max} = 1727 (5) keV. It is consistent with the adopted maximum energy of 1732.18 (49) keV deduced from the Q-value [2021Wa16].

Table 2 – Measured β^+ endpoint energy of ^{15}O decay.

Reference	E_{max} (keV)	Comment
1934Li04	1700	No uncertainty; excluded.
1936Fowl	1700 (200)	
1937Step	1660 (300)	Excluded.
1950Br29	1638 (5)	Excluded
1955Ki28	1735 (8)	
1957Ki22	1726 (5)	Weighted mean and minimum experimental uncertainty of the two reported values: 1723 (5) keV and 1736 (10) keV.
1963Va31	1700 (20)	
Evaluated	1727 (5)	Weighted mean with minimum experimental uncertainty
Adopted	**1732.18 (49)**	**From AME2020 [2021Wa16]**

2 Atomic data

The fluorescence yield ω_K has been taken from Schönfeld *et al.* [1996Sc06].

3 Gamma emission

The annihilation radiation emission probability, without the correction factor for the annihilation-in-flight process in the medium, has been calculated by Saisinuc [2008DuZX] from the related decay data as ($2 \times P_{\beta+}$): $I_{\gamma\pm}$ = 199.8002 (34) %.

4 References

1934Li04 M.S. Livingston, E. McMillan, Phys. Rev. 46, 437 (1934) [Half-life, E_{max}]

1935Mc01 E.M. McMillan, M.S. Livingston, Phys. Rev. 47, 452 (1935) [Half-life]

1936Fowl W.A. Fowler, L.A. Delsasso, C.C. Lauritsen, Phys. Rev. 49, 561 (1936) [E_{max}]

1937Step W.E. Stephens, K. Djanab, T.W. Bonner, Phys. Rev. 52, 1079 (1937) [E_{max}]

1950Br29 H. Brown, V. Perez-Mendez, Phys. Rev. 78, 649 (1950) [Half-life, E_{max}]

1954Kl36 R.M. Kline, D.J. Zaffarano, Phys. Rev. 96, 1620 (1954) [Half-life]

1955Ba83 S. Bashkin, R.R. Carlson, E.B. Nelson, Phys. Rev. 99, 107 (1955) [Half-life]

1955Ki28 J.D. Kington, J.K. Bair, H.O. Cohn, H.B. Willard, Phys. Rev. 99, 1393 (1955) [E_{max}]

1957Ki22 O.C. Kistner, A. Schwarzschild, B.M. Rustad, D.E. Alburger, Phys. Rev. 105, 1339 (1957) [Half-life, E_{max}]

1957Pe12 J.R. Penning, F.H. Schmidt, Phys. Rev. 105, 647 (1957) [Half-life]

1959Ki99 O.C. Kistner, B.M. Rustad, Phys. Rev. 114, 1329 (1959) [Half-life]

1960Ja12 J. Janecke, Z. Naturforsch. 15a, 593 (1960) [Half-life]

1963Cs02 J. Csikai, G. Peto, Phys. Letters 4, 252 (1963) [Half-life]

1963Ne05 J.W. Nelson, E.B. Carter, G.E. Mitchell, R.H. Davis, Phys. Rev. 129, 1723 (1963) [Half-life]

1963Va31 S.S. Vasilev, L.Y. Shavtvalov, Izv. Akad. Nauk SSSR, Ser. Fiz. 27, 1261 (1963); Bull. Acad. Sci. USSR, Phys. Ser. 27, 1239 (1964) [Half-life, E_{max}]

1972Le33 W. Leiper, R.W.P. Drever, Phys. Rev. C6, 1132 (1972) [K/β^+]

1977Az01 G. Azuelos, J.E. Kitching, K. Ramavataram, Phys. Rev. C15, 1847 (1977) [Half-life]

1977Ba48 W. Bambynek, H. Behrens, M.H. Chen, B. Crasemann, M.L. Fitzpatrick, K.W.D. Ledingham, H. Genz, M. Mutterer, R.L. Intemann, Rev. Mod. Phys. 49, 77 (1977); Erratum Rev. Mod. Phys. 49, 961 (1977) [Theory, compilation]

1991Aj01 F. Ajzenberg-Selove, Nucl. Phys. A 523, 1 (1991) [Evaluation]

1996Sc06 E. Schönfeld, H. Janssen, Nucl. Instrum. Methods Phys. Res. A369, 527 (1996) [Atomic data]

2008DuZX C. Dulieu, M.-M. Bé, V. Chisté, Proc. Intern. Conf. Nuclear Data for Science and Technology p.97 (2008) [Saisinuc]

2019Mo35 X. Mougeot, Appl. Rad. Isotopes 154 (2019), 108884 [BetaShape program]

2020Bu02 D.P. Burdette, M. Brodeur, D.W. Bardayan, F.D. Becchetti, D. Blankstein, C. Boomershine, L. Caves, S.L. Henderson, J.J. Kolata, B. Liu, J. Long, P.D. O'Malley, S.Y. Strauss, Phys. Rev. C 101, 055504 (2020) [Half-life]

2021Wa16 M. Wang, W.J. Huang, F.G. Kondev, G. Audi, S. Naimi, Chin. Phys. C 45, 030003 (2021) [Q-value]

2023Mo21 X. Mougeot, Appl. Radiat. Isot. 201, 111018 (2023) [BetaShape program]

2023Se01 N. Severijns, L. Hayen, V. De Leebeeck, S. Vanlangendonck, K. Bodek, D. Rozpedzik, I.S. Towner, Phys. Rev. C 107, 015502 (2023) [Theory, compilation]